

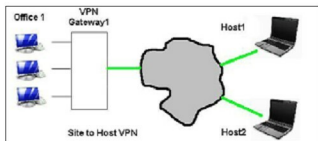
The logo for IPCop OPEN VPN. It features a green circular icon with a black silhouette of a person wearing a hard hat. To the right of the icon, the text "IPCop" is in green, and "OPEN VPN" is in black.

Part—4

The first article in this series, in the December 2011 issue, discussed the basic installation and configuration of IPCop 1.4.21. Subsequent articles discussed five important add-ons used to enhance IPCop's functionality: Advanced Proxy, URL Filter, Update Accelerator, Calamaris and Block Outgoing Traffic. Here, we will look at how to extend IPCop to incorporate Site to Host OpenVPN, using the Zerina add-on.

Before we begin, let us try to understand some of the basics about Virtual Private Networks (VPNs). There are two basic types of VPNs:

1. **Transport mode VPN:** Consider one remote host (notebook) that requires to communicate with the office network using a public Internet connection. To ensure confidentiality and integrity of the communication, all traffic must be encrypted. This functionality is provided by a VPN gateway, typically configured on a firewall. Transport mode VPN is primarily used to establish secure host-to-network communication (and sometimes between two hosts). Refer to Diagram 1. In a transport mode VPN, the host initiates the connection to the VPN gateway, negotiates various protocols and establishes a secure connection. The connecting external hosts are aware that traffic to the office network is through a VPN.
2. **Tunnel mode VPN:** IPsec VPN tunnel mode is primarily used to establish connections between two sites, though it can also be used for host-to-network connections. We'll discuss more about IPsec VPN tunnels in another article.



Site to Host VPN

```

echo -e "\nChecking current version of IPEX:"
echo -e "\n1. Looking for older versions of IPEX: \n"
echo -e "\n2. Checking for valid IPEX-version: \n"

if [ -x "$(command -v grep)" ]; then
    echo
    echo "ERROR: This IPEX edition only works with IPEX 1.4.15" \
    echo "Sorry, Please update your IPEX - visit www.ipex.org/en"
    echo ""
    exit 1
fi
echo -e "\nIPEX OK!"

```

Figure 1: Patching the Zerina Installer for 1.4.21 support

GNU implementation of VPN

OpenVPN is open source VPN software. And as described at www.openvpn.net, it enables you to:

- tunnel any IP sub-network or virtual Ethernet adapter over a single UDP or TCP port;
- configure a scalable, load-balanced VPN server farm using one or more machines, which can handle thousands of dynamic connections from incoming VPN clients;
- use all of the encryption, authentication and certification features of the OpenSSL library to protect your private network traffic as it transits the Internet.

OpenVPN is available in repositories of most FOSS OSs, such as Debian, Ubuntu, Fedora, FreeBSD and Maemo. The OpenVPN client is available for MS Windows, Mac OS and Android (it works on rooted devices).

With the basics covered, let's move on to Zerina!

Zerina

The Zerina add-on adds an OpenVPN server functionality to IPCop. Let's set it up and access the office network using a Windows-based client.

Download the latest Zerina binaries from openvpn.eu; SCP them to the IPCop box; log in via SSH (don't forget: TCP port 222); and then extract the Zerina *gzip* file in a new folder. Note that these binaries were developed long ago for IPCop 1.4.15. To

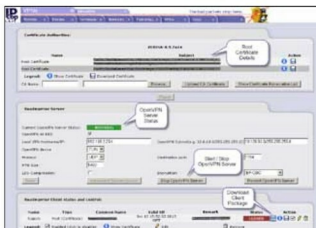


Figure 2: OpenVPN server has started